

First Order ODEs - Exact equations

- Differential of Function of two variables: $z = f(x, y)$*

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

- For the special case: $z = f(x, y) = c$*

$$dz = 0 \Rightarrow \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = 0 \Rightarrow M(x, y)dx + N(x, y) dy = 0$$

$$\text{where, } M(x, y) = \frac{\partial z}{\partial x} \text{ and } N(x, y) = \frac{\partial z}{\partial y}$$

$$\text{Then, } \frac{\partial M}{\partial y} = \frac{\partial^2 z}{\partial y \partial x} \text{ and } \frac{\partial N}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}. \quad [\text{If } M(x, y) \text{ and } N(x, y) \text{ are continuous and have first partials}]$$

$$\text{Thus, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

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Definition. A first-order differential equation of the form

$$M(x, y)dx + N(x, y)dy = 0$$

is called an **exact differential equation** if $M(x, y)dx + N(x, y)dy$ is **exactly** the total differential of $f(x, y)$.

Criterion for an Exact ODEs. If $M(x, y)$, $N(x, y)$ are continuous in x and y , and have continuous first partial derivatives, then a necessary and sufficient condition that

$$M(x, y)dx + N(x, y)dy = 0$$

be an exact differential equation is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

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Example. Solve the differential equations,

$$(x + y) dx + (x - y)dy = 0 \quad [\text{The ODE is not separable}]$$

Solution. Here, $M(x, y) = x + y$ and $N(x, y) = x - y$ where

$$\frac{\partial M}{\partial y} = 1, \quad \frac{\partial N}{\partial x} = 1 \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

STEP-01
Exactness
Test

Thus, the given equation is an exact and there exists a solution $f(x, y) = c$ such that

$$\left\{ \begin{array}{l} \frac{\partial f}{\partial x} = M(x, y) = x + y, \quad \text{and} \quad \frac{\partial f}{\partial y} = N(x, y) = x - y \\ \Rightarrow \int \frac{\partial f}{\partial x} dx = \int (x + y) dx \quad \Rightarrow x + g'(y) = x - y \Rightarrow g'(y) = -y \\ \Rightarrow f(x, y) = \frac{x^2}{2} + xy + g(y) \quad \Rightarrow \int g'(y) dy = - \int y dy \\ \Rightarrow g(y) = -\frac{y^2}{2} \end{array} \right.$$

STEP-02
Find $f(x, y)$

STEP-03
Find $g(y)$

STEP-04
Write the Solution

The desired solution becomes, $f(x, y) = c \Rightarrow \frac{x^2}{2} + xy - \frac{y^2}{2} = c \Rightarrow x^2 + 2xy - y^2 = c$

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Example. Solve the differential equations,

$$2xy \, dx + (x^2 - 1)dy = 0 \quad [\text{The ODE is separable}]$$

Solution. Here, $M(x, y) = 2xy$ and $N(x, y) = x^2 - 1$ where

$$\frac{\partial M}{\partial y} = 2x, \text{ and } \frac{\partial N}{\partial x} = 2x \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus, the given equation is an exact and there exists a solution $f(x, y) = c$ such that

$$\begin{aligned} \frac{\partial f}{\partial x} &= M(x, y) = 2xy, & \text{and} & & \frac{\partial f}{\partial y} &= N(x, y) = x^2 - 1 \\ \Rightarrow \int \frac{\partial f}{\partial x} dx &= \int 2xy \, dx & & & \Rightarrow x^2 + g'(y) &= x^2 - 1 \Rightarrow g'(y) = -1 \\ \Rightarrow f(x, y) &= x^2y + g(y) & & & \Rightarrow \int g'(y) \, dy &= - \int dy \\ & & & & \Rightarrow g(y) &= -y \end{aligned}$$

The desired solution becomes, $f(x, y) = c \Rightarrow x^2y - y = c$

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Example. Solve the differential equations,

$$(e^{2y} - y \cos xy) dx + (2xe^{2y} - x \cos xy + 2y)dy = 0 \quad [\text{The ODE is not separable}]$$

Solution. Here, $M(x, y) = e^{2y} - y \cos xy$ and $N(x, y) = 2xe^{2y} - x \cos xy + 2y$ where

$$\frac{\partial M}{\partial y} = 2e^{2y} - \cos xy + xy \sin xy, \text{ and } \frac{\partial N}{\partial x} = 2e^{2y} - \cos xy + xy \sin xy \Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Thus, the given equation is an exact and there exists a solution $f(x, y) = c$ such that

$$\begin{aligned} \frac{\partial f}{\partial x} &= M(x, y) = e^{2y} - y \cos xy, & \text{and} & & \frac{\partial f}{\partial y} &= N(x, y) = 2xe^{2y} - x \cos xy + 2y \\ \Rightarrow \int \frac{\partial f}{\partial x} dx &= \int (e^{2y} - y \cos xy) dx & & & \Rightarrow 2xe^{2y} - x \cos xy + g'(y) &= 2xe^{2y} \\ & & & & & & -x \cos xy + 2y \\ \Rightarrow f(x, y) &= xe^{2y} - \sin xy + g(y) & & & \Rightarrow g'(y) = 2y &\Rightarrow \int g'(y) dy = \int 2y dy \\ & & & & & & \Rightarrow g(y) = y^2 \end{aligned}$$

The desired solution becomes, $f(x, y) = c \Rightarrow xe^{2y} - \sin xy + y^2 = c$

First Order ODEs - Exact equations

H.W. from the text book

Exercises 2.4

Determine whether the given differential equation is exact. If it is exact, solve it.

1. $(2x - 1) dx + (3y + 7) dy = 0$

2. $(2x + y) dx - (x + 6y) dy = 0$

3. $(5x + 4y) dx + (4x - 8y^3) dy = 0$

4. $(\sin y - y \sin x) dx + (\cos x + x \cos y - y) dy = 0$

5. $(2xy^2 - 3) dx + (2x^2y + 4) dy = 0$

6. $\left(2y - \frac{1}{x} + \cos 3x\right) \frac{dy}{dx} + \frac{y}{x^2} - 4x^3 + 3y \sin 3x = 0$

7. $(x^2 - y^2) dx + (x^2 - 2xy) dy = 0$

8. $\left(1 + \ln x + \frac{y}{x}\right) dx = (1 - \ln x) dy$

9. $(x - y^3 + y^2 \sin x) dx = (3xy^2 + 2y \cos x) dy$

10. $(x^3 + y^3) dx + 3xy^2 dy = 0$

11. $(y \ln y - e^{-xy}) dx + \left(\frac{1}{y} + x \ln y\right) dy = 0$

12. $(3x^2y + e^y) dx + (x^3 + xe^y - 2y) dy = 0$

13. $x \frac{dy}{dx} = 2xe^x - y + 6x^2$

14. $\left(1 - \frac{3}{y} + x\right) \frac{dy}{dx} + y = \frac{3}{x} - 1$

15. $\left(x^2y^3 - \frac{1}{1 + 9x^2}\right) \frac{dx}{dy} + x^3y^2 = 0$

16. $(5y - 2x)y' - 2y = 0$

17. $(\tan x - \sin x \sin y) dx + \cos x \cos y dy = 0$

18. $(2y \sin x \cos x - y + 2y^2 e^{xy^2}) dx$
 $= (x - \sin^2 x - 4xye^{xy^2}) dy$

19. $(4t^3y - 15t^2 - y) dt + (t^4 + 3y^2 - t) dy = 0$

20. $\left(\frac{1}{t} + \frac{1}{t^2} - \frac{y}{t^2 + y^2}\right) dt + \left(ye^y + \frac{t}{t^2 + y^2}\right) dy = 0$

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Exercises 2.4

H.W. from the text book

Solve the given initial-value problem.

21. $(x + y)^2 dx + (2xy + x^2 - 1) dy = 0, \quad y(1) = 1$

22. $(e^x + y) dx + (2 + x + ye^y) dy = 0, \quad y(0) = 1$

23. $(4y + 2t - 5) dt + (6y + 4t - 1) dy = 0, \quad y(-1) = 2$

24. $\left(\frac{3y^2 - t^2}{y^5} \right) \frac{dy}{dt} + \frac{t}{2y^4} = 0, \quad y(1) = 1$

25. $(y^2 \cos x - 3x^2y - 2x) dx$
 $+ (2y \sin x - x^3 + \ln y) dy = 0, \quad y(0) = e$

26. $\left(\frac{1}{1 + y^2} + \cos x - 2xy \right) \frac{dy}{dx} = y(y + \sin x), \quad y(0) = 1$

Verify that the given differential equation is not exact. Multiply the given differential equation by the indicated integrating factor $\mu(x, y)$ and verify that the new equation is exact. Solve.

29. $(-xy \sin x + 2y \cos x) dx + 2x \cos x dy = 0;$
 $\mu(x, y) = xy$

30. $(x^2 + 2xy - y^2) dx + (y^2 + 2xy - x^2) dy = 0;$
 $\mu(x, y) = (x + y)^{-2}$